



Tianjiao Yan (闫天娇)

Peking University

School of Life Sciences

Biological Sciences Major

Undergraduate Class of 2026

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PERSONAL STATEMENT

Third-year undergraduate student majoring in Biological Sciences at Peking University with interdisciplinary background in biology and computer science. Possesses solid data analysis capabilities, rapid learning ability, and excellent communication and collaboration skills. Self-taught multiple programming languages while studying biological sciences, with front-end development experience. Focuses on AI cutting-edge fields with extensive experience in large language model usage and prompt optimization, as well as experience in AI-assisted programming, language model tool development, and Agent framework usage and optimization.

EDUCATION

• Peking University

Expected July 2026

B.S. in Biological Sciences, School of Life Sciences

Beijing, China

- Relevant Coursework: Neurobiological Basis of Learning and Memory, Physiology, Biochemistry
- Academic Performance: Strong academic record with focus on neuroscience and computational biology

• Yangcun No.1 High School, Wuqing District

Sept 2019 - July 2022

High School Diploma

Tianjin, China

- Academic Performance: Top 1% of graduating class
- Demonstrated excellence in mathematics and sciences

RESEARCH EXPERIENCE

• Hippocampal Neural Circuit Research Project

Jan 2023 - Sept 2023

Research Assistant, Supervisor: Prof. Chenglin Miao

- Investigated spatial cognition using VR technology and reward-based stimulation while monitoring brain electrical activity to explore inter-regional coordination patterns.
- Utilized water maze paradigms to simulate low/zero gravity conditions and analyze brain electrical discharge patterns in mice under altered gravitational environments.
- Operated electrophysiological recording equipment and imaging systems to collect neural signals and behavioral data under various experimental conditions.

• Peking University AI Agent Project

Mar 2025 - Present

Front-end Developer

- Leading front-end development of China's first multi-user AI knowledge platform using Vue, Nuxt and WebSocket technologies.
- Implemented real-time multi-user conversation interfaces with large language model integration and Agent mode execution capabilities.

• Anaerobic Microbial Ecology Research Project

Apr 2024 - Oct 2024

Research Assistant

- Independently conducted anaerobic microbial ecology laboratory research project, learning and mastering complete anaerobic microbial cultivation techniques and operational procedures from scratch.
- Successfully completed and became proficient in anaerobic methanogen culture medium preparation, anaerobic cultivation, and bacterial strain passage procedures.
- Reviewed and analyzed relevant academic literature, designed and validated experiments investigating changes in oxygen tolerance characteristics of anaerobic methanogens under specific antioxidant conditions.

LEADERSHIP & PROJECT EXPERIENCE

• Peking University AI Agent Project

Mar 2025 - Present

Front-end Developer

- Participating in Peking University Computing Center's AI agent development initiative.
- Leading the design and front-end development of China's first multi-user AI knowledge platform - Xiaobei Intelligent Space.
- Proficiently utilizing Vue, Nuxt, and related front-end technologies to build multi-user real-time conversation interfaces integrated with large language model interaction capabilities, along with data visualization and permission management features.
- Implementing front-end logic and partial back-end logic for domestic large language model API integration, MCP tool invocation, and Agent mode execution. Innovatively achieved real-time incremental synchronization of inference results based on WebSocket, resolving user experience issues in existing systems under multi-terminal, multi-user scenarios.

• Biomedical Industry Field Research

Aug 2023

Research Team Member

- Participated in biomedical industry field research project, collaborating with team members to develop research plans and establish connections with multiple biomedical companies including Fanngjing Bio, Tuoxinda Bio, and WuXi AppTec for in-depth interviews and site visits.
- Gained comprehensive understanding of biomedical industry operational models, supply chain components, competitive landscape, and development trends, with focused research on differentiated development strategies between startup and large-scale biomedical enterprises.
- Analyzed the "high investment, long return cycle" characteristics of the biomedical industry, gaining insights into small biomedical companies' "project team-based" operational models and customer-oriented market strategies.

• Student Union Department of Practice

2023-2024

Director

- Comprehensively managed department operations, leading team members in various practical activities while enhancing department cohesion and execution capabilities.
- Spearheaded planning and organization of college "Reading Club" series activities and "Ideological Practice Field Investigation" activities, coordinating 7 practice teams and organizing over 100 students for nationwide themed social practice research.

TECHNICAL SKILLS & COMPETENCIES

Programming Languages: Python, R, JavaScript with front-end development and large language model project experience

Data Analysis Tools: Proficient in Python/R for data analysis, data mining, and data visualization

Laboratory Skills: Electrophysiological recording, anaerobic microbial cultivation, behavioral analysis, VR-based experimental design

Software Proficiency: Office suite, Adobe Creative Suite (PS/AI/PR), statistical analysis software

AI & Machine Learning: Large language model usage and optimization, prompt engineering, Agent framework development

Languages: Native Chinese, English (CET-6), strong reading and writing capabilities in academic English

AWARDS & HONORS

- **Third Prize, Peking University High Performance Computing Competition** *2025*
- **University-level Scholarship - Social Service Award, Peking University** *2023*
- **"Rising Star in Student Organizations" - PKU Communist Youth League Committee** *2023*